

Mastering Mathematical Formulae

A Complete Guide to Substitution & Rearrangement

Math’s Coursebook Edition — Designed for Easy Learning (Ages 10+)

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1 Introduction: What is a Formula?

A **formula** is a magic rule that we write down once so that we never have to work the same kind of problem out from scratch again. We put in the numbers we know, and the formula gives us back the number we want.

1.1 Real-Life Example: The Taxi Ride

Imagine a taxi that charges Rs 250 just to start your journey, plus Rs 45 for every kilometer you travel. We can write this as a simple formula!

If we let n be the number of kilometers traveled, and F be the total fare, the formula becomes:

$$F = 250 + 45n$$

For a 17 km trip, we simply replace the letter n with 17:

$$F = 250 + 45 \times 17 = 250 + 765 = 1015$$

So, the fare is Rs 1015. The beauty of this is that the same rule now works for any distance, without doing the whole calculation from scratch again!

1.2 The Basic Toolkit

To master formulae, you need to know three key tools:

- **Substitute:** Replace each letter with its given number, then work out the answer following the rules of **BODMAS** (Brackets, Orders, Division/Multiplication, Addition/Subtraction).
- **Subject:** The letter standing completely alone on one side of the equals sign. In $F = 250 + 45n$, the letter F is the subject.
- **Changing the Subject:** Do the same opposite operation to both sides of the equals sign until the letter you want stands completely alone.

1.3 Common Formulae Table

Here are some of the most useful formulae that you will use in math and real life:

Quantity	Formula	What it means
Area of a rectangle	$A = l \times w$	Area = length $\times$ width
Perimeter of a rectangle	$P = 2(l + w)$	Perimeter = $2 \times$ (length + width)
Speed	$s = d \div t$	Speed = distance $\div$ time
Cost with a fixed charge	$C = F + r \times n$	Cost = Fixed charge + (rate $\times$ number of items)

Math's Coursebook Study Advice

Subtracting a negative number is exactly the same as adding a positive number! For example:
 $-(\frac{1}{2} \times -24) = +12$.

Math's Coursebook Study Advice

To clear fractions easily, multiply every term by the Lowest Common Multiple (LCM) of the denominators. To undo a square, take the square root at the very end.

2 The 8 Learning Cases

2.1 Case 1 — Substituting Values

The Situation: Replace each letter with the number you are given, then work the answer out following the order of operations (BODMAS). Keep the units in mind, because the answer takes its units from the formula.

Worked Example

Example 1: The area of a rectangle is $A = l \times w$. Find A when $l = 12$ cm and $w = 7$ cm.

Solution: Put the numbers in place of the letters and multiply:

$$A = 12 \times 7 = 84 \text{ cm}^2 \quad \text{[2 marks]}$$

Worked Example

Example 2: A formula for the time needed to cook a pie is $t = 50m + 25$, where t is the number of minutes and m is the mass in kilograms. For how long, in hours and minutes, should a pie of mass 2.5 kg be cooked?

Solution:

1. Substitute $m = 2.5$ into the formula: $t = 50 \times 2.5 + 25$
2. Multiply before you add, as BODMAS requires: $t = 125 + 25 = 150$ minutes [2 marks]
3. The question asks for hours and minutes, so convert the answer: 150 minutes = 2 hours 30 minutes [1 mark]

Math's Coursebook Study Advice

Substitute into the formula exactly as it is written. Replace the letter, keep every operation just as it was, and then work through with BODMAS. The answer carries the units of the formula, and those units may need converting afterwards (e.g., 150 minutes becomes 2 hours 30 minutes).

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. $A = l \times w$. Find A when $l = 8$ and $w = 5$.
2. $P = 2(l + w)$. Find P when $l = 10$ and $w = 4$.
3. $s = d \div t$. Find s when $d = 120$ and $t = 3$.
4. $C = 5n + 100$. Find C when $n = 20$.

2.2 Case 2 — Negatives and Powers

The Situation: Substituting a negative number, or a number that is then squared, is where most marks are lost in this topic. Writing the number inside brackets keeps its sign safe.

Watch Out!

When $x = -5$, the value of x^2 is **not** -25 . We write it as:

$$(-5)^2 = -5 \times -5 = 25$$

because a negative number multiplied by itself always gives a positive answer!

Worked Example

Example 1: Evaluate $2a - b^2$ when $a = 5$ and $b = 3$, and $a^2 - b$ when $a = -2$ and $b = 3$.

Solution:

$$(a) \quad 2(5) - (3)^2 = 10 - 9 = 1 \quad [2 \text{ marks}]$$

$$(b) \quad (-2)^2 - 3 = 4 - 3 = 1 \quad [2 \text{ marks}]$$

Worked Example

Example 2: Use the formula $a = 5b - \frac{1}{2}c$ to find a when:

(a) $b = 4$ and $c = 12$

$$a = 5 \times 4 - \left(\frac{1}{2} \times 12\right) = 20 - 6 = 14 \quad [2 \text{ marks}]$$

(b) $b = -2$ and $c = -24$

Keep each part in its own bracket so no sign is lost:

$$a = (5 \times -2) - \left(\frac{1}{2} \times -24\right) = -10 - (-12)$$

Subtracting a negative turns into adding a positive:

$$a = -10 + 12 = 2 \quad [2 \text{ marks}]$$

Math's Coursebook Study Advice

Subtracting a negative is the same as adding a positive: $-10 - (-12) = -10 + 12 = 2$.

Also, watch where the bracket sits! $2p^2$ means $2 \times p \times p$, but $(2p)^2$ means $2p \times 2p = 4p^2$, because the bracket squares the 2 as well!

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. Evaluate $3a - 2$ when $a = 4$.
2. Evaluate x^2 when $x = -5$.
3. Evaluate $2x^2$ when $x = 3$.
4. Evaluate $a^2 - b$ when $a = -2$ and $b = 3$.

2.3 Case 3 — Writing a Formula from Words

The Situation: Turn a description into a formula by giving each quantity a letter, and then translating words like "per", "plus" and "times" into math symbols.

- "A fixed charge plus a rate per item" becomes $C = \text{fixed} + \text{rate} \times n$.
- "Multiplied by 4 and 3 added" becomes $y = 4n + 3$.

Worked Example

Example 1: A plumber charges a Rs 500 call-out fee plus Rs 200 per hour. Write a formula for the cost C of h hours.

Solution: The call-out fee is paid once (fixed), and the Rs 200 is paid for each hour. So:

$$C = 500 + 200h \quad \text{[2 marks]}$$

Worked Example

Example 2: Write each word equation using only the given letters and numbers:

(a) y is 3 more than 4 times the value of x

$$y = 4x + 3 \quad \text{[1 mark]}$$

(b) b is twice the value of 5 more than c

$$b = 2(c + 5) \quad \text{[1 mark]}$$

(c) m is half the value of 6 less than n

$$m = (n - 6) \div 2 \quad \text{[1 mark]}$$

Math's Coursebook Study Advice

When a whole expression is doubled or halved, it needs brackets!

"Twice the value of 5 more than c " is written as $b = 2(c + 5)$, and not $2c + 5$.

"Half the value of 6 less than n " halves the whole of $n - 6$, so $m = (n - 6) \div 2$.

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. A number n is multiplied by 4 and 3 is added. Write a formula for the result y .
2. A plumber charges Rs 500 plus Rs 200 per hour h . Write a formula for the cost C .
3. Write a formula for the perimeter P of a square of side s .
4. A phone plan is Rs 300 fixed plus Rs 2 per minute m . Write a formula for the bill B .

2.4 Case 4 — Brackets and Fractions

The Situation: Many formulae contain brackets and fractions. Work out the top and the bottom of a fraction — or whatever is inside the brackets — first, and only then combine the two.

Worked Example

Example 1: Evaluate $(a + b) \div 2$ when $a = 6$ and $b = 10$, and $3(x - 2)$ when $x = 5$.

Solution:

(a) $(6 + 10) \div 2 = 16 \div 2 = 8$ [1 mark]

(b) $3(5 - 2) = 3 \times 3 = 9$ [1 mark]

Worked Example

Example 2: Use the formula $x = (y^2 - z^2) \div (5w)$ to find x when $y = 9$, $z = 1$ and $w = 2$.

Solution:

1. Work out the top (numerator) on its own: $y^2 - z^2 = 9^2 - 1^2 = 81 - 1 = 80$ [1 mark]

2. Now work out the bottom (denominator) on its own: $5w = 5 \times 2 = 10$ [1 mark]

3. Only now divide one by the other: $x = 80 \div 10 = 8$ [1 mark]

Math's Coursebook Study Advice

Fractions and order of operations: When the whole formula is one fraction, work out the whole of the top and the whole of the bottom separately, and divide only at the end.

Remember: a fraction bar acts like a giant bracket. So $(6x - 8) \div 2$ divides both terms to give $3x - 4$.

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. Evaluate $(a + b) \div 2$ when $a = 6$ and $b = 10$.
2. Evaluate $3(x - 2)$ when $x = 5$.
3. Evaluate $(a + b + c) \div 3$ when $a = 4$, $b = 5$ and $c = 6$.
4. Evaluate $v^2 \div 2$ when $v = 4$.

2.5 Case 5 — Changing the Subject

The Situation: To make a different letter the subject, undo the operations around it one at a time, always doing the same thing to both sides of the equals sign. This is exactly the same process as solving an equation!

- $y = x + 5$: subtract 5 from both sides to get $x = y - 5$.
- $v = u + at$: subtract u , then divide by t , to get $a = (v - u) \div t$.

Worked Example

Example 1: Make a the subject of $v = u + at$.

Solution:

1. Subtract u from both sides: $v - u = at$.
2. Now divide both sides by t :

$$a = (v - u) \div t \quad \text{[2 marks]}$$

Worked Example

Example 2: Make s the subject of $p = \frac{1}{2}q - \frac{1}{3}r + \frac{1}{4}s$, then find s when $p = 5$, $q = 1$ and $r = 6$.

Solution:

1. The LCM of 2, 3 and 4 is 12, so multiply every single term by 12:

$$12p = 6q - 4r + 3s \quad \text{[1 mark]}$$

2. Move the other terms across, then divide by 3:

$$3s = 12p - 6q + 4r \implies s = (12p - 6q + 4r) \div 3 \quad \text{[2 marks]}$$

3. Now substitute the given values:

$$s = (12 \times 5 - 6 \times 1 + 4 \times 6) \div 3 = (60 - 6 + 24) \div 3 = 78 \div 3 = 26 \quad \text{[1 mark]}$$

Math's Coursebook Study Advice

If the formula contains fractions, clear them first by multiplying every term by the Lowest Common Multiple (LCM) of the denominators. There is then nothing left to trip over! The subject must end up written exactly once, on its own, on one side of the equals sign.

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. Make x the subject: $y = x + 5$.
2. Make x the subject: $y = 3x$.
3. Make a the subject: $v = u + at$.
4. Make r the subject: $C = 2\pi r$.

2.6 Case 6 — Simplify First, Then Change the Subject

The Situation: Some formulae arrive with "like terms" sitting on one or both sides. Collect those terms first, because the rearranging that follows is then only one short step!

Worked Example

Example: By collecting like terms, simplify $4y - 3y = 5x + 2x + 11 - 8$ and make y the subject.

Solution:

1. Take the left-hand side first: $4y - 3y = y$.
2. Now the right-hand side: $5x + 2x = 7x$ and $11 - 8 = 3$.
3. The letter y is already alone, so the work is finished:

$$y = 7x + 3 \quad \text{[2 marks]}$$

Math's Coursebook Study Advice

Collect the like terms on each side separately before you touch the equals sign at all. Remember that $4y - 3y$ is simply y , because a coefficient of 1 is never written down!

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Your Turn

1. Simplify and make y the subject: $5y - 2y = 6x + 3x + 10 - 4$.
2. Simplify and make m the subject: $9m - 4m = 10n + 5n - 20$.
3. Simplify and make t the subject: $4t + 2t = 8s - 3s + 12$.
4. Simplify and make p the subject: $7p - 6p + 2 = 5q + 3q - 1$.

2.7 Case 7 — Rearrange First, Then Substitute

The Situation: When the letter you need is not the subject, rearrange the whole formula first and put the numbers in afterwards. Substituting first leaves you solving an equation with awkward numbers in it.

Worked Example

Example: (a) Rearrange $2y + 6x = 9$ to express y in terms of x . (b) Find the value of y when $x = -2$.

Solution:

(a) Subtract $6x$ from both sides:

$$2y = 9 - 6x$$

Now divide both sides by 2. Here the decimal is exact, so either form is acceptable:

$$y = (9 - 6x) \div 2 \quad \text{or} \quad y = 4.5 - 3x \quad \textbf{[2 marks]}$$

(b) Substitute $x = -2$ into the rearranged formula:

$$y = 4.5 - (3 \times -2) = 4.5 + 6 = 10.5 \quad \textbf{[2 marks]}$$

Math's Coursebook Study Advice

Always make the wanted term positive before doing anything else! So $8 - \frac{1}{3}y = 2x$ becomes $8 = 2x + \frac{1}{3}y$.

Clear a fractional coefficient by multiplying both sides by the denominator: $\frac{1}{3}y = 8 - 2x \implies y = 24 - 6x$.

A decimal may be used in a rearranged formula only if it is exact. If it is not, keep the fraction instead.

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. Rearrange $2y = 11 - 6x$ to make y the subject, then find y when $x = 1$.
2. Rearrange $2p - 8q = 15$ to make p the subject, then find p when $q = 0.5$.
3. Rearrange $\frac{1}{2}j + 7 = 8k$ to make j the subject, then find j when $k = -1$.
4. Rearrange $2n - 3m = 4$ to express m in terms of n , then find m when $n = 5$.

2.8 Case 8 — Squares and Square Roots in Formulae

The Situation: When the letter you want is squared, or sits under a square root sign, make the square (or the root) the subject first and undo it at the very last step.

Worked Example

Example: The acceleration of a jet, a m/s², is given by $a = (v^2 - u^2) \div (2s)$, where u and v are the initial and final velocities in m/s and s is the distance in metres.

(a) Find a when $u = 3$, $v = 7$ and $s = 4$.

$$a = (7^2 - 3^2) \div (2 \times 4) = (49 - 9) \div 8 = 40 \div 8 = 5 \text{ m/s}^2 \quad [2 \text{ marks}]$$

(b) Make v^2 the subject, then find v when $u = 9$, $a = 4$ and $s = 5$.

Multiply both sides by $2s$ to clear the fraction:

$$v^2 - u^2 = 2as \implies v^2 = u^2 + 2as \quad [1 \text{ mark}]$$

Substitute the values into that rearranged formula:

$$v^2 = 9^2 + (2 \times 4 \times 5) = 81 + 40 = 121 \quad [1 \text{ mark}]$$

Take the square root only at the very end:

$$v = \sqrt{121} = 11 \text{ m/s} \quad [1 \text{ mark}]$$

Math's Coursebook Study Advice

Take one step at a time! Make v^2 the subject before you try to make v the subject.

Squaring is undone by taking the square root, and a square root is undone by squaring. Unless you are told otherwise, always give the positive root.

Remember: $c^2 = a^2 + b^2$ gives $c = \sqrt{a^2 + b^2}$, which is **not** the same as $a + b$!

This advice is brought to you by **Math's Coursebook**.

Your Turn

1. Make b^2 the subject of $c^2 = a^2 + b^2$.
2. Using $c^2 = a^2 + b^2$, find the positive value of a when $c = 13$ and $b = 5$.
3. Using $v^2 = u^2 + 2as$, find v when $u = 8$, $a = 3$ and $s = 6$.
4. Use $a = (v^2 - u^2) \div (2s)$ to find a when $u = 2$, $v = 8$ and $s = 5$.

3 The Question Bank

Work through these questions in order to build your confidence and mastery!

3.1 Level 1: Foundation Questions (1–28)

- 1) $A = lw$. Find A when $l = 6$ and $w = 4$.
- 2) $P = 4s$. Find P when $s = 5$.
- 3) $y = 2x + 1$. Find y when $x = 3$.
- 4) $C = 10n$. Find C when $n = 7$.
- 5) $s = d \div t$. Find s when $d = 100$ and $t = 5$.
- 6) Evaluate $3a$ when $a = 6$.
- 7) Evaluate $a + b$ when $a = 5$ and $b = 8$.
- 8) $V = lwh$. Find V when $l = 2$, $w = 3$ and $h = 4$.
- 9) Evaluate $2x - 3$ when $x = 5$.
- 10) $P = 2(l + w)$. Find P when $l = 6$ and $w = 2$.
- 11) $y = x^2$. Find y when $x = 4$.
- 12) Evaluate $5n + 2$ when $n = 10$.
- 13) A number is doubled. Write a formula for the result y from n .
- 14) Evaluate $a - b$ when $a = 3$ and $b = 7$.
- 15) $C = 50 + 5n$. Find C when $n = 6$.
- 16) Evaluate $x \div 2$ when $x = 18$.
- 17) Use the formula $y = 23 - x$ to find y when $x = -7$. *[Math's Coursebook]*
- 18) Use the formula $y = 20 - x$ to find y when $x = -5$. *[Math's Coursebook]*
- 19) Use the formula $y = 13 + 3x$ to find y when $x = -4$. *[Math's Coursebook]*
- 20) Use the formula $p = 9 - \frac{1}{2}q$ to find p when $q = -8$. *[Math's Coursebook]*
- 21) Use the formula $w = 3x - 2y$ to find w when $x = -2$ and $y = -4$. *[Math's Coursebook]*
- 22) Rearrange $q = p - 5$ to make p the subject. *[Math's Coursebook]*
- 23) p is 7 less than 3 times the value of q . Write a formula for p . *[Math's Coursebook]*
- 24) j is 2 more than half the value of k . Write a formula for j . *[Math's Coursebook]*
- 25) A pie takes $t = 50m + 25$ minutes to cook. Find t when $m = 1.5$. *[Math's Coursebook]*
- 26) The velocity of a rocket is $v = u + at$. Find v when $u = 100$, $a = 4$ and $t = 25$. *[Math's Coursebook]*
- 27) Use the formula $y^2 = 16 - x^2$ to find y^2 when $x = 1$. *[Math's Coursebook]*
- 28) Use the formula $c^2 = a^2 + b^2$ to find the positive value of c when $a = 5$ and $b = 12$. *[Math's Coursebook]*

3.2 Level 2: Intermediate Questions (29–56)

- 29) Evaluate $3a - 2$ when $a = 4$.
- 30) Evaluate x^2 when $x = -5$.
- 31) Evaluate $a^2 - b$ when $a = -2$ and $b = 3$.
- 32) Taxi fare: $F = 250 + 45n$. Find F when $n = 17$.
- 33) Evaluate $(a + b) \div 2$ when $a = 6$ and $b = 10$.
- 34) Make x the subject: $y = x + 5$.
- 35) Make a the subject: $v = u + at$.
- 36) Evaluate $2x^2$ when $x = 3$.
- 37) Write a formula: the cost C is Rs 500 plus Rs 200 per hour h .
- 38) Evaluate $3(x - 2)$ when $x = 5$.
- 39) $V = \pi r^2 h$. Find V to the nearest whole number when $r = 2$ and $h = 5$, taking $\pi = 3.14$.
- 40) Make x the subject: $y = 3x$.
- 41) Evaluate $a^2 + 2a$ when $a = -3$.
- 42) Make r the subject: $C = 2\pi r$.
- 43) Evaluate $(a + b + c) \div 3$ when $a = 4$, $b = 5$ and $c = 6$.

- 44) Evaluate $4x - x^2$ when $x = 3$.
- 45) Use the formula $j = \frac{1}{2}k - \frac{1}{3}m$ to find j when $k = -10$ and $m = -24$. *[Math's Coursebook]*
- 46) b is twice the value of 5 more than c . Write a formula for b . *[Math's Coursebook]*
- 47) m is half the value of 6 less than n . Write a formula for m . *[Math's Coursebook]*
- 48) Rearrange $13 + 5b = 15c$ to express b in terms of c . *[Math's Coursebook]*
- 49) Find the value of b when $c = 2.8$, given that $13 + 5b = 15c$. *[Math's Coursebook]*
- 50) Use the formula $p = (q + 8) \div (q - 2)$ to find p when $q = 6$. *[Math's Coursebook]*
- 51) Find the value of s when $t = 5$, given that $s = (t^2 + 3) \div (t - 3)$. *[Math's Coursebook]*
- 52) Use the formula $x = z^3 - y^2$ to find x when $z = 3$ and $y = 4$. *[Math's Coursebook]*
- 53) Evaluate q when $r = 16$ and $p = -2$, using the formula $q = \sqrt{r} + p^2$. *[Math's Coursebook]*
- 54) Use the formula $x = (y^2 - z^2) \div (5w)$ to find x when $y = 13$, $z = 7$ and $w = 6$. *[Math's Coursebook]*
- 55) Rearrange $x^2 + y^2 = 16$ to make y^2 the subject, and find y^2 when $x = -3$. *[Math's Coursebook]*
- 56) Using $y^2 = 16 - x^2$, find the positive value of y when $x = 3.5$, correct to 2 decimal places. *[Math's Coursebook]*

3.3 Level 3: Challenge Questions (57–78)

- 57) The area of a trapezium is $A = \frac{1}{2}(a + b)h$. Find A when $a = 6$, $b = 10$ and $h = 4$.
- 58) Make h the subject of $A = \frac{1}{2}(a + b)h$.
- 59) Evaluate $s = ut + \frac{1}{2}at^2$ when $u = 5$, $t = 2$ and $a = 10$.
- 60) Make u the subject of $v = u + at$.
- 61) A party costs Rs 5000 plus Rs 350 per guest. Write a formula, and find the cost for 40 guests.
- 62) Evaluate $a^2 + b^2$ when $a = 3$ and $b = 4$, then take the square root.
- 63) Make x the subject of $y = 2x - 7$.
- 64) $F = 9C \div 5 + 32$ converts Celsius to Fahrenheit. Find F when $C = 20$.
- 65) Make C the subject of $F = 9C \div 5 + 32$.
- 66) Evaluate $(3a - b) \div (a + b)$ when $a = 4$ and $b = 2$.
- 67) Simplify and make p the subject: $9q - 5q + 7 = 12p - 11p + 1$. *[Math's Coursebook]*
- 68) Simplify and make z the subject: $\frac{3}{2}z - 3 = 5w + \frac{1}{2}z$. *[Math's Coursebook]*
- 69) The total surface area of a solid cone is $A = \pi r(r + l)$. Find A in terms of π when $r = 4$ cm and $l = 8$ cm. *[Math's Coursebook]*
- 70) Make l the subject of $A = \pi r^2 + \pi rl$. *[Math's Coursebook]*
- 71) Make t the subject of $v = u + at$, then find t when $v = 300$, $u = 60$ and $a = 5$. *[Math's Coursebook]*
- 72) A pie of mass m kg needs $t = 50m + 25$ minutes of cooking. Calculate the mass of a pie that should be cooked for 1 hour and 55 minutes. *[Math's Coursebook]*
- 73) The sum of all the integers from 1 to n is $(n^2 + n) \div 2$. Write down a formula for the sum of all the integers from 4 to n . *[Math's Coursebook]*
- 74) Find the sum of all the integers from 4 to 200. *[Math's Coursebook]*
- 75) Given that $t + 3 = 10 - \sqrt{2s}$, find the value of t when $s = 32$. *[Math's Coursebook]*
- 76) Use the formula $c = \sqrt{d - e^3}$ to evaluate c when $d = 16900$ and $e = 5$. *[Math's Coursebook]*
- 77) Use the formula $c = \sqrt{d - e^3}$ to evaluate c when $d = 900$ and $e = d \div 300$. *[Math's Coursebook]*
- 78) A rectangle 16 cm long and 11 cm high has a rectangle $3p$ cm by p cm removed from it. Find a formula for the shaded area A , and find A when $p = 4$. *[Math's Coursebook]*

3.4 Level 4: Exam-Style Questions (Paper 1, Non-Calculator) (79–96)

- 79)** $A = lw$. Find A when $l = 9$ and $w = 6$. [1]
- 80)** $y = 3x - 4$. Find y when $x = 5$. [1]
- 81)** A taxi charges Rs 200 plus Rs 40 per km. Write a formula for the fare F for n km. [1]
- 82)** Evaluate $2x^2 + 1$ when $x = -3$. [2]
- 83)** Make x the subject of $y = x - 6$. [1]
- 84)** Make a the subject of $v = u + at$. [2]
- 85)** $P = 2(l + w)$. Find P when $l = 12$ and $w = 7$. [2]
- 86)** The cost C of hiring a hall is Rs 3000 plus Rs 150 per hour. Find the cost for 5 hours. [2]
- 87)** Evaluate $s = \frac{1}{2}at^2$ when $a = 10$ and $t = 3$. [2]
- 88)** Make r the subject of $A = \pi r^2$. [2]
- 89)** Use the formula $a = 5b - \frac{1}{2}c$ to find a when $b = -2$ and $c = -24$. [Math's Coursebook] [2]
- 90)** Rearrange $2y + 6x = 9$ to express y in terms of x . [Math's Coursebook] [2]
- 91)** Make y the subject of $8 - \frac{1}{3}y = 2x$. [Math's Coursebook] [2]
- 92)** The acceleration of a jet is $a = (v^2 - u^2) \div (2s)$. Find a when $u = 3$, $v = 7$ and $s = 4$. [Math's Coursebook] [2]
- 93)** Make v^2 the subject of $a = (v^2 - u^2) \div (2s)$. [Math's Coursebook] [2]
- 94)** The height of a solid cylinder is $h = (A - 2\pi r^2) \div (2\pi r)$. Find h when $A = 264\pi$ and $r = 6$. [Math's Coursebook] [3]
- 95)** The circumference and area of a circle are connected by $c = 2\sqrt{\pi A}$. Find c in terms of π when $A = 9\pi$. [Math's Coursebook] [2]
- 96)** A rectangle 11 cm by 9 cm has a rectangle 8 cm by x cm cut from it. Write a formula for the shaded area A , and find A when $x = 3$. [Math's Coursebook] [3]

4 Complete Answer Key

4.1 Answers to the “Your Turn” Exercises

Case	Answers
Case 1	1) 40 2) 28 3) 40 4) 200
Case 2	1) 10 2) 25 3) 18 4) 1
Case 3	1) $y = 4n + 3$ 2) $C = 500 + 200h$ 3) $P = 4s$ 4) $B = 300 + 2m$
Case 4	1) 8 2) 9 3) 5 4) 8
Case 5	1) $x = y - 5$ 2) $x = y \div 3$ 3) $a = (v - u) \div t$ 4) $r = C \div (2\pi)$
Case 6	1) $y = 3x + 2$ 2) $m = 3n - 4$ 3) $t = (5s + 12) \div 6$ 4) $p = 8q - 3$
Case 7	1) $y = (11 - 6x) \div 2$; $y = 2.5$ 2) $p = (15 + 8q) \div 2$; $p = 9.5$ 3) $j = 16k - 14$; $j = -30$ 4) $m = (2n - 4) \div 3$; $m = 2$
Case 8	1) $b^2 = c^2 - a^2$ 2) $a = 12$ 3) $v = 10$ 4) $a = 6$

4.2 Answers to the Question Bank

• Foundation (1–28):

- | | |
|--------------|----------------------------|
| 1) 24 | 15) 80 |
| 2) 20 | 16) 9 |
| 3) 7 | 17) 30 |
| 4) 70 | 18) 25 |
| 5) 20 | 19) 1 |
| 6) 18 | 20) 13 |
| 7) 13 | 21) 2 |
| 8) 24 | 22) $p = q + 5$ |
| 9) 7 | 23) $p = 3q - 7$ |
| 10) 16 | 24) $j = \frac{1}{2}k + 2$ |
| 11) 16 | 25) 100 |
| 12) 52 | 26) 200 |
| 13) $y = 2n$ | 27) 15 |
| 14) -4 | 28) 13 |

• Intermediate (29–56):

- | | |
|--------------------------|----------------------------------|
| 29) 10 | 43) 5 |
| 30) 25 | 44) 3 |
| 31) 1 | 45) 3 |
| 32) 1015 | 46) $b = 2(c + 5)$ |
| 33) 8 | 47) $m = (n - 6) \div 2$ |
| 34) $x = y - 5$ | 48) $b = (15c - 13) \div 5$ |
| 35) $a = (v - u) \div t$ | 49) 5.8 |
| 36) 18 | 50) 3.5 |
| 37) $C = 500 + 200h$ | 51) 14 |
| 38) 9 | 52) 11 |
| 39) 63 (62.8) | 53) 8 |
| 40) $x = y \div 3$ | 54) 4 |
| 41) 3 | 55) $y^2 = 16 - x^2$; $y^2 = 7$ |
| 42) $r = C \div (2\pi)$ | 56) 1.94 |

• Challenge (57–78):

57) 32

58) $h = 2A \div (a + b)$

59) 30

60) $u = v - at$

61) $C = 5000 + 350g$; Rs 19 000

62) 5

63) $x = (y + 7) \div 2$

64) 68

65) $C = 5(F - 32) \div 9$

66) $5/3$

67) $p = 4q + 6$

68) $z = 5w + 3$

69) $48\pi \text{ cm}^2$

70) $l = (A - \pi r^2) \div (\pi r)$

71) $t = (v - u) \div a$; $t = 48$

72) 1.8 kg

73) $(n^2 + n) \div 2 - 6$

74) 20094

75) -1

76) 5

77) 3

78) $A = 176 - 3p^2$; $A = 128 \text{ cm}^2$

• Exam-Style (79–96):

79) 54

80) 11

81) $F = 200 + 40n$

82) 19

83) $x = y + 6$

84) $a = (v - u) \div t$

85) 38

86) Rs 3750

87) 45

88) $r = \sqrt{A \div \pi}$

89) 2

90) $y = (9 - 6x) \div 2$ or $y = 4.5 - 3x$

91) $y = 24 - 6x$

92) 5 m/s^2

93) $v^2 = u^2 + 2as$

94) 16

95) 6π

96) $A = 99 - 8x$; $A = 75 \text{ cm}^2$